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Sentiment Analysis of COVID-19 Vaccine on Twitter Social Media: Systematic Literature Review

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***Abstract*-This study attempted to review studies on sentiment analysis of COVID-19 vaccine using Twitter as the source data. It was conducted to understand methods to collect, preprocess and classify data by researchers. We used Systematic Literature Review approach to collect, filter and review research papers found. Research papers were collected from IEEE, ScienceDirect, Springer, ACM Digital Library, arXiv, medRxiv and Google Scholar by taking studies published from 2020 until June 2021. First of all, we gathered paper based on title and it resulted forty papers. On the next step, the contents of the papers were examined using exclusion criteria and inclusion criteria to investigate them further. This step filtered out several papers leaving twenty one corresponding papers to be reviewed for this study. We reviewed the remaining papers to answer four predetermined research questions. The frequently used methods to collect data are RTweet, Twint, Twitter API and Tweepy. Several techniques which used by researchers to preprocess Twitter data are stop word removal, remove punctuation and link, case folding, tokenization, stemming, remove duplicate tweet and lemmatization. To classify sentiment tweets, researchers used several machine learning and deep learning methods. BERT as Transformers method was also used by several researchers. Further studies regarding methods and variables or parameters to classify tweet data are still needed.**

Keywords—Twitter, COVID-19 vaccine, sentiment analysis, positive sentiment, negative sentiment

I. INTRODUCTION

The COVID-19 virus has spread throughout the world since 2019. Based on data from Worldometer in August 2021, there have been more than 206 million cases of COVID with a death rate of more than 4 million patients. Therefore, researchers are starting to conduct research in order to develop vaccines for COVID-19. Several vaccines which have been approved by WHO are: Pfizer, AstraZeneca and Moderna, while several other vaccines are in the process of review and assessment by World Health Organization (WHO) [1].

The vaccination process has been initiated by the United States and Israel since December 2020 and was followed by other countries, including Indonesia [2]. The vaccination process has caused controversy in several countries. The results of the study [3] revealed that there are several countries that have low public acceptance of vaccines. This needs to be followed up to reduce the spread of COVID-19 globally.

It is necessary to know in advance about community perceptions regarding implementation of the vaccination policy. One technique which can be addressed to understand public perceptions of vaccination is sentiment analysis. Sentiment analysis uses Natural Language Processing (NLP), a text analysis and computation technique, to extract and classify sentiments [4]. Many studies have analyzed sentiment on the COVID-19 problem with data from various social media. One of the data sources on social media that is often used to conduct sentiment analysis research is Twitter.

This study aims to portray commonly used methods to collect, preprocess and classify Twitter data toward COVID-19 vaccination. This study also intends to get a depiction of sentiment data from several studies which analyzed sentiment data from Twitter.

Several surveys have been done before, such as study [5] conducted a review of 24 selected studies that identified Twitter's role during the COVID-19 pandemic. The results of their study described topics discussed by Twitter users during the pandemic and analysis of the emotions behind user tweets. It reviewed of research that conducted studies on the advantages and disadvantages of using Twitter during the COVID-19 pandemic as well.

The survey conducted by the author [6] provided an overview of open source data which can be used by future researchers. This study described a complete description of the data sources both in terms of data types, applications, methods and from the repository side. Sentiment analysis in this study was categorized using text data types that came from Twitter and were applied using NLP.

Study [7] identified the use of Twitter in times of pandemics and natural disasters. This study classified the research found based on the type of disaster and the research writing theme. From this research, it is recommended that countries or organizations prevent the spread of false information by disseminating true and accurate information from the organization using official accounts of social media. It was also revealed that analysis of data on Twitter is also a good source of data to be able to understand the mental state of the community, estimate the number of people injured in a disaster and model the distribution of an outbreak.

The studies above are still unable to provide a complete overview of the sentiment analysis research on Twitter, especially regarding public sentiment towards the COVID-19 vaccine. In this study, we used SLR method to conduct paper review to answer this problem.

II. METHODS

Systematic Literature Review (SLR) is a method of collecting research results from various sources to answer predetermined research questions [8]. Although initially SLRs were used in the health sector, nowadays SLR is also widely used in the engineering field. This study used

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the SLR approach in reviewing research papers. The author [8] stated that there are six steps in conducting SLR which listed below.

A. Research Question

RQ1 : What techniques are used to perform data collection?
RQ2 : What techniques are used in data preprocessing?
RQ3 : What methods are used in conducting sentiment analysis and classification?
RQ4 : How are the results of the sentiment analysis of the research being reviewed?

B. Search Strategy

We searched papers from IEEE, ScienceDirect, Springer, ACM Digital Library, arXiv, medRxiv and Google Schoar with two keyword combinations, for Indonesian and English.:

- "sentiment analysis" and "twitter" and ("COVID vaccine" or "vaccine hesitancy" or "vaccine rejection" or "vaccine challenge").
- "analisis sentimen" and "twitter" and ("vaksin COVID" or "penolakan vaksin" or "tantangan vaksin").

C. Study Selection

In conducting paper screening, it is necessary to determine the screening criteria. Author [8] explained that there are two kinds of criteria that can be used in screening papers, inclusion criteria and exclusion criteria. Inclusion criteria are criteria that make a research paper included in the list of research being reviewed, while the exclusion criteria are criteria that make a research paper not included in the list of research being reviewed. The inclusion criteria in this study are as follows:

- The researches collected were researches published from 2020 to June 2021.
- Papers are written in Indonesian or English.
- The research topic has to be sentiment analysis or sentiment classification on the COVID-19 vaccine.
- The data source used were from Twitter.

While the exclusion criteria in this study are as follows:

- Research that does not meet the inclusion criteria.
- The study did not clearly state the method used.
- Research that could not answer research objective.

Search process and number of identified studies at each stage are shown in Fig. 1. In the first stage, the selection which was based on title and abstract resulted 40 studies. In the next stage, 21 papers were selected as full article reviews.

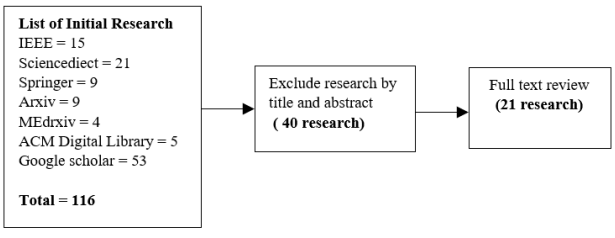


Fig. 1 Selection Process

D. Quality Assessment

Quality assessment is needed to maintain the quality of the paper being reviewed. One way to do this is by reviewing only high quality papers. Due to the limited number of papers that discuss the sentiment analysis of the COVID-19 vaccine on social media twitter, it was quite difficult to do a quality assessment.

E. Data Extraction

At this stage, the data taken from the reviewed paper were year of publication, languages, data set used and data collection technique, data preprocessing techniques, methods used in conducting sentiment analysis and classification, as well as the results of sentiment analysis. The data obtained from the data extraction process were collected in the form of a spreadsheet document.

F. Data Synthesis

At this stage, paper selection was done by considering whether the paper can help answer the research question and meet the inclusion and exclusion criteria. In this study, forty journals were obtained and after applying the research questions, inclusions and criteria resulted in twenty one papers which were finally reviewed and analyzed.

III. RESULT AND DISCUSSION

There were forty studies which passed the first stage of selection for review of the abstract and the content. The second stage of selection resulted in twenty one studies. These studies were reviewed to explore the five topics to be discussed in this study, namely data collection, data preprocessing classification and detection of topics, as well as the results of sentiment analysis on each of them.

A. Data Collection Technique

In general, there are two types of sentiment analysis research based on data collection technique. The first type is analysis sentiment research which uses public dataset and the second type is research which performs data collection. Authors [6] reviewed various public datasets, both in image, textual and sound format.

There are several data collection technique used by previous researchers. For example, web scraping with RTweet technique by authors [9] [10], [11] and [12]. It was published by [13] to make it easier for R language users to collect data from Twitter. The R Tweet package was designed to simplify the data collection process and make interactions with the Twitter API more accessible to many users [14].

Data collection can also be performed using several tools. For example, Twint Scraping Tools which was implemented in studies [15], [16] [17] and [18]. When using the tool, users do not need to use the Twitter API which has API restrictions. Another tool is Mozdeh which was used by authors [19] Mozdeh was created by the Statistical Cybermetrics Research Group at the University of Wolverhampton. The next tool is Netlytic, a software that performs queries against the Twitter Rest API [20]. This tool was used by authors [21]. Crowdbreak platform, a digital project using Twitter's stream filter API to capture

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a set of keywords related to COVID-19 in English was used in studies [22] and [23]. Authors [24], [25] and [26] imported Tweepy as library in their Python program so it can communicate with the Twitter API. Tweepy's documentation can be accessed at [27]. Data collection techniques are shown in Fig. 2 with the number of research.

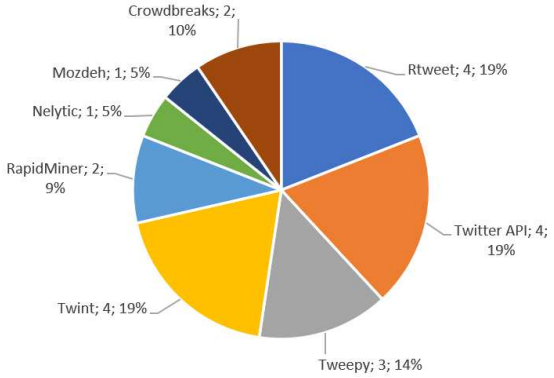


Fig. 2 Data Collection Technique

The last tool is Rapid Miner which was applied by authors [28] and [29]. Rapid Miner can handle various problems that exist in Data Mining such as classification, clustering and association [30]. In practice, Rapid Miner uses an access token that must be owned by user to gain access to Twitter before crawling and extracting data from Twitter [28].

In researches [20], [31],[32] and [33] data collection did not use the tools as mentioned above, but was done by writing the source code itself with the Twitter API.

B. Data Pre-processing

Preprocessing data is an important step in conducting sentiment analysis, considering that the data from Twitter in an unstructured form. Unstructured data cannot be processed directly by the classification algorithm because of the nature of the algorithm which generally only accepts input in the form of number. Algorithms will also face the difficulty to understand variety of writing and language styles and even the way each person writes a tweet.

The initial stage in data preprocessing is data cleaning. At data cleaning stage, unnecessary characters, words, phrases and even tweets are removed from the tweet. For example, studies [9], [17], [19], [22], [23], [28] and [29] deleted on duplicate tweets.

Other parts of tweets which are usually also removed from tweet data are URLs or links, hashtags, mentions and special characters (punctuation) as was performed in [9], [10], [11], [15], [16], [17], [18], [21], [20], [22], [24], [26], [31], [32] and [33]. This deletion was done because they do not have direct relevance to a person's sentiments on a topic of conversation or discussion. Furthermore, studies [9], [10], [11], [12], [15], [16], [17], [18], [21], [20], [25], [28], [29], [31], [32] and [33] also removed stop words from tweets. Elimination of stop words was done by referencing to a dictionary containing a list of stop words. It should be noted that some researchers have different opinion about the word "no" and "not", whether these two words are a stop word or not because in analytical sentiment, these two words can change a sentiment to the opposite.

After cleaning up the data, generally all words in the tweet data are changed to a lowercase case as in [9], [10], [11], [12], [15], [26], [31] [32] and [33] in order to make all tweet data are in the same form or case. Thus, even though a word was written in a different way by different user, in the end it still has the same meaning. In addition, studies [15], [16], [17], [18], [20] and [25] restored a word to its original form (lemma). This technique is known as lemmatization. Fig. 3 shows the number of studies that use pre-processing technique.

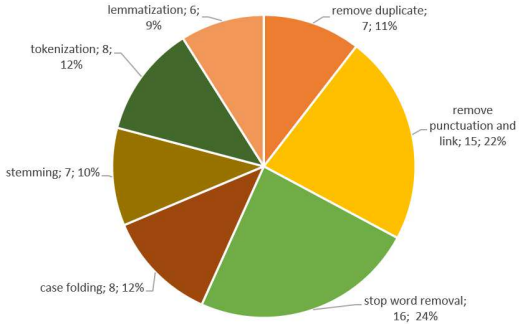


Fig. 3 Data Preprocessing Technique

Another technique that is also used in NLP is Tokenization. Tokenization is a process to break a sentence into word pieces before becoming input for the model [34]. Tokenization was used in studies [21], [15], [17], [24], [26], [28], [29] and [33]. Furthermore, authors [9] normalized the dataset, so that words which were not formal and did not fit spelling rules were converted into formal words.

In researches [11], [12], [16], [18], [26], [28] and [29] the Stemming technique was used. Stemming has the same goal as Lemmatization, which is to return a word to its original form [35]. Although they have the same purpose, these two techniques have different way of returning the word to its original form. Lemmatization uses a dictionary to find out the original form of a word. This technique is very useful when used in a language that uses many affixes such as Indonesian. Mistakes may occur when the cut is done on a word which should not have cut. One example of a stemming algorithm for Indonesia is Nazief & Andriani algorithm [36].

NLP allows feature extraction of data. One technique which commonly used is the Bag of Word (BoW) as was performed in studies [11] and [15]. Each word will be given a score considering the number of words and their frequency. Word Hashing [37] and Term Frequency - Inverse Document Frequency (TF-IDF) [38] are methods which can score each word. The TF-IDF technique was used in studies [11] and [29].

C. Classification and Topic Detection

After completing the data preprocessing stages, there are two possible works to do which are classification and topic detection.

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1. Classification

Generally, there are three classification methods used in classifying tweets regarding the COVID 19 vaccine, namely machine learning, deep learning and transformers. Machine learning methods such as Random Forest (RF), Support Vector Machine (SVM), Multilayer Perceptron (MLP), Gradient Boosting (GB), Naïve Bayes (NB) and Linear Regression (LR) were used in researches [11], [20], [25], [26] and [31]. Machine learning methods are usually combined with other methods such as CBOW and Skip Gram as in study [11], Unigram, Bi-gram and FastText methods as in study [20] and N-gram as in research [31]. Authors [25] made a variety of Support Vector Machine and Naïve Bayes in their study. Table 1 provides an overview of the methods which can classify tweets in sentiment analysis.

TABLE 1. OVERVIEW OF THE METHODS CLASSIFICATION

Research	Method Classification
[11], [31]	Random Forest (RF)
[11], [20], [26], [31]	Support Vector Machine (SVM)
[11]	Multilayer Perceptron (MLP)
[11]	Gradien Boosting (GB)
[20], [26], [28],[29], [31]	Naïve Bayes (NB)
[20]	Logistic Regresion (LR)
[11]	LSTM
[31], [25]	Bi-LSTM
[31]	CNN
[20], [22], [23], [25], [31], [32]	BERT

We found three deep learning methods which were used in selected papers. The first method is Long-Short Term Memory (LSTM) as in study [11]. The second method is Bidirectional LSTM (Bi-LSTM) as in studies [31] and [25]. The last method is Convolutional Neural Network (CNN) which was used in study [31].

The Bidirectional Encoder Representations from Transformers (BERT) method is a Transformer-based machine learning method specifically for NLP developed by Jacob and his team from Google in 2018 [39][40]. This method is used in studies [20], [22], [23], [25], [31] and [32] and. Their studies found that the accuracy of BERT is higher than other methods.

Deep learning methods have many parameters and each author used different configuration in creating model. This study is still unable to depict this issue. Furthermore, BERT has many variants which also not yet described in this study.

2. Topic Detection

Researches [9], [12], [21], [15], and [33] not only conducted a sentiment analysis of tweets, but also explored the most discussed topic by users. Topic detection was conducted using the Latent Dirichlet Allocation (LDA) method [41]. The LDA method was originally used in the context of population genetic in 2000. In 2003, the method began to be used in machine learning by [13]. LDA method is considered as a sub-problem of NLP which purpose is to determine a topic automatically from a sentence.

D. Sentiment Analysis Results

Sixteen of twenty one studies classified the results of the sentiment analysis with a positive-negative label or other similar classification way. Two studies stated that the percentage of negative sentiment is higher than positive sentiment. They were research [28] which studied sentiment in Indonesian and research [20] which studied sentiment in English and Portuguese. Summarizes the result of reviewed papers in this study shown on Table 2.

Other studies presented research results in different ways. For example, research [21] did not present sentiment data in the form of a percentage, but in the form of the number of sentiments juxtaposed with the time dimension.

TABLE 2. SUMMARY OF RESEARCH RESULTS FROM REVIEWED PAPERS

Research	Study Result
[9], [10], [11], [12], [16], [17], [18], [22], [24], [25], [26], [29], [31], [32], [33], [23]	Positive sentiment > negative sentiment
[20], [28]	Positive sentiment < negative sentiment
[21]	Visualization of sentiment data juxtaposed with time dimension
[15], [19]	Topic detection for negative sentiment

Research [15] conducted topic detection only on tweets with negative sentiments with the aim of knowing the causes of negative public sentiment towards the COVID-19 vaccine. The same thing was done in research [19], but the labelling of topics was done manually. This is still possible because the number of tweets is not too large. The research data reviewed can be found in Table 3.

TABLE 3. REVIEWED PAPER

Year	Study	Language	Data Collection	Preprocessing	Method
2020	[9]	Indonesian	Rtweet	Delete duplicate, remove punctuation, remove link, stop word removal, case folding.	Lexicon for classification, LDA to find topics.
2021	[10]	India	Rtweet	Remove White Spaces, Links, remove punctuation, stop words removal, case folding	NRC Emoticon Lexicon.
2020	[11]	English	Rtweet	Remove punctuation, stop word removal, case folding, stemming, remove white space	TF-IDF, n-gram, BoW. ML : RF, SVM, MLP, GB, LSTM.
2021	[12]	English	Rtweet	Remove non-English words, stop word removal, case folding, and stemming	LDA
2020	[15]	English	Twint	Remove link, case folding, remove white space, tokenization, lemmatization, stop word removal	LDA

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Year	Study	Language	Data Collection	Preprocessing	Method
2021	[16]	English	Twint	Stop word removal, remove number, remove punctuation, remove link, stemming, lemmatization	LDA to find topics and TextBlob for analysis sentiment
2021	[17]	English	Twint	Delete duplicate, remove non-english, stop word removal, remove link, tokenization, lemmatization	VADER & TextBlob
2021	[18]	English	Twint	Remove punctuation, remove link, remove number, stop word removal, stemming, lemmatization	TextBlob
2021	[19]	English	Mozdeh	Delete duplicate	Cohen's Kappa
2021	[21]	Multi languange	Netlytic	Remove punctuation, stop word removal, tokenization.	LDA to find topics and TextBlob for analysis sentiment
2021	[20]	English, Portuguese	Twitter API	Remove punctuation, remove link, remove number, stop word removal, lemmatization	Machine Learning: NB; LR; RF; SVM and BERT
2020	[22]	English	Crowdbreaks	Delete duplicate, remove punctuation	BERT
2020	[23]	English	Crowdbreaks	Delete duplicate	FastText & BERT
2021	[24]	English	Tweepy	Remove punctuation, remove link, tokenization	TextBlob
2021	[25]	English	Tweepy	Remove non-English words, stop word removal, lemmatization	BERT + Bi-LSTM
2021	[26]	Indonesian	Tweepy	Case folding, remove punctuation, tokenizing, stop word removal, normalization, and stemming.	NB; SVM
2020	[28]	Indonesian	Rapid Miner & Drone Emprit	Delete duplicate, stop word removal, tokenization, stemming	NB
2021	[29]	English, Tagalog	Rapid Miner	Delete duplicate, stop word removal, tokenization, stemming	NB
2021	[31]	English	Twitter Filtered Stream API	Stop word removal, remove white space, case folding, remove link, remove punctuation,	MNB, RF, SVM, Bi-LSTM, CNN and BERT
2020	[32]	English	Twitter API	Remove link, stop word removal	BERT
2020	[33]	English	Twitter API	Remove punctuation, stop word removal, tokenization, case folding	VADER, LDA

IV. CONCLUSION AND FUTURE DIRECTIONS

This study summarizes various sentiment analysis studies on the COVID-19 vaccine, which data source are from Twitter. After reviewing the title, abstract and content of the research found on IEEE, ScienceDirect, Springer, ACM Digital Library, arXiv and medRxiv, twenty one studies were selected for further review. The review was done in terms of data collection techniques, data preprocessing, methods of classification and topic detection, and as well as result of sentiment analysis research.

There are several tools that can be used for collecting data from Twitter. The tools which frequently used are RTweet, Twint, Tweepy and Twitter API. After the data has been collected, data need to be prepared in data cleaning and data preprocessing process. Data cleaning process includes removing duplicate tweets, deleting, links, hashtags, mentions, punctuation and stop words. Data preprocessing was performed using lowercase, lemmatization, tokenization, stemming and feature extraction techniques. For studies which use data in Bahasa, other authors commonly used Nazief & Andriani algorithm for stemming, although other algorithms are also possible to implemented.

Machine learning methods such as Random Forest, Support Vector Machine, Multilayer Perceptron, Gradient Boosting, Naïve Bayes and Logistic Regression are capable of classifying tweet data. In deep learning, methods such as LSTM, Bidirectional LSTM and CNN were also used in several studies for classification. Lastly, the transformers method, BERT is also capable of doing classification task and were the most used method compared to other methods in this review. In addition, some researchers also detect topics in tweet data using the LDA method.

Most of the sentiment analysis researches were conducted in English. Meanwhile, sentiment analysis on other languages is still relatively few. The result of the reviewed researches shows that the public's positive sentiment towards the COVID-19 vaccine is greater than negative sentiment. The cause of negative sentiment can be found by detecting the topic in the negative sentiment tweet.

This research still has not reviewed the classification method in detail, thus further studies are still needed to provide a comprehensive overview of the method and the variables or parameters used to readers who want to learn about sentiment analysis using data from Twitter. Further studies also need to be done considering that research on sentiment analysis will continue to emerge from other researchers from various countries as vaccinations are implemented around the world and the development of the COVID-19 vaccine in each country.

Lastly, we still could not find public twitter dataset about COVID-19 vaccine in Bahasa. This can be a chance for the next authors to explore and contribute.

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